Higher-Feature Formulas

Dependency Table

Higher feature	Depends on base features	Count
Z_{P_T}	$p_{T,\ell_1},\ \eta_{\ell_1},\ \phi_{\ell_1},\ p_{T,\ell_2},\ \eta_{\ell_2},\ \phi_{\ell_2}$	6
Z_η	$p_{T,\ell_1},\ \eta_{\ell_1},\ \phi_{\ell_1},\ p_{T,\ell_2},\ \eta_{\ell_2},\ \phi_{\ell_2}$	6
Z_Φ	$p_{T,\ell_1},\ \eta_{\ell_1},\ \phi_{\ell_1},\ p_{T,\ell_2},\ \eta_{\ell_2},\ \phi_{\ell_2}$	6
Z_{Mass}	$p_{T,\ell_1},\ \eta_{\ell_1},\ \phi_{\ell_1},\ p_{T,\ell_2},\ \eta_{\ell_2},\ \phi_{\ell_2}$	6
$Z_{\Delta R}$	$\eta_{\ell_1},\ \phi_{\ell_1},\ \eta_{\ell_2},\ \phi_{\ell_2}$	4
W_{P_T}	$p_{T,\ell},\ \phi_\ell,\ p_{T,\mathrm{miss}},\ \phi_{\mathrm{miss}}$	4
W_Φ	$p_{T,\ell},\ \phi_\ell,\ p_{T,\mathrm{miss}},\ \phi_{\mathrm{miss}}$	4
W_{M_T}	$p_{T,\ell},\ \phi_\ell,\ p_{T,\mathrm{miss}},\ \phi_{\mathrm{miss}}$	4
Top_{P_T}	$p_{T,\ell},\ \phi_\ell,\ p_{T,\mathrm{miss}},\ \phi_{\mathrm{miss}},\ p_{T,b},\ \phi_b$	6
Top_Φ	$p_{T,\ell},\ \phi_\ell,\ p_{T,\mathrm{miss}},\ \phi_{\mathrm{miss}},\ p_{T,b},\ \phi_b$	6
Top_{M_T}	$p_{T,\ell},\ \phi_\ell,\ p_{T,\mathrm{miss}},\ \phi_{\mathrm{miss}},\ p_{T,b},\ \phi_b$	6

$$p_x = p_T \cos\phi, \qquad p_y = p_T \sin\phi, \qquad p_z = p_T \sinh\eta$$
$$E = \sqrt{p_x^2 + p_y^2 + p_z^2}$$

Z from two leptons ℓ_1, ℓ_2

$$\vec{p}_Z = \vec{p}_{\ell_1} + \vec{p}_{\ell_2}, \qquad E_Z = E_{\ell_1} + E_{\ell_2}$$

$$Z_{P_T} = \sqrt{p_{Zx}^2 + p_{Zy}^2}, \qquad Z_\Phi = \mathrm{atan2}(p_{Zy}, p_{Zx}), \qquad Z_\eta = \mathrm{artanh}\left(\frac{p_{Zz}}{\sqrt{p_{Zx}^2 + p_{Zy}^2 + p_{Zz}^2}}\right)$$

$$Z_{\mathrm{Mass}} = \sqrt{\max\Big(E_Z^2 - \Big(p_{Zx}^2 + p_{Zy}^2 + p_{Zz}^2\Big), 0\Big)}$$

(equivalent for massless leptons) $\qquad Z_{\mathrm{Mass}}^2 = 2\, p_{T,\ell_1} p_{T,\ell_2} \left(\cosh(\Delta\eta_{12}) - \cos(\Delta\phi_{12})\right)$

$$\Delta\phi_{12} = \mathrm{atan2}(\sin(\phi_1 - \phi_2), \cos(\phi_1 - \phi_2)), \qquad Z_{\Delta R} = \sqrt{(\eta_1 - \eta_2)^2 + \Delta\phi_{12}^2}$$

W (transverse) from lepton + MET

$$p_{Wx} = p_{\ell x} + p_{\mathrm{miss},x}, \qquad p_{Wy} = p_{\ell y} + p_{\mathrm{miss},y}$$
$$W_{P_T} = \sqrt{p_{Wx}^2 + p_{Wy}^2}, \qquad W_\Phi = \mathrm{atan2}(p_{Wy}, p_{Wx})$$
$$\Delta\phi_{\ell\nu} = \mathrm{atan2}(\sin(\phi_\ell - \phi_{\mathrm{miss}}), \cos(\phi_\ell - \phi_{\mathrm{miss}}))$$
$$W_{M_T} = \sqrt{\max(2\, p_{T,\ell}\, p_{T,\mathrm{miss}}\, (1 - \cos\Delta\phi_{\ell\nu}), 0)}$$

Top (transverse proxy) from W + b-jet

$$p_{tx} = p_{Wx} + p_{bx}, \qquad p_{ty} = p_{Wy} + p_{by}$$

$$\mathrm{Top}_{P_T} = \sqrt{p_{tx}^2 + p_{ty}^2}, \qquad \mathrm{Top}_{\Phi} = \mathrm{atan2}(p_{ty}, p_{tx})$$

$$\Delta\phi_{Wb} = \mathrm{atan2}(\sin(\phi_W - \phi_b), \cos(\phi_W - \phi_b))$$

$$\mathrm{Top}_{M_T} = \sqrt{\max((W_{M_T} + p_{T,b})^2 + 2\, W_{P_T}\, p_{T,b}\, (1 - \cos \Delta\phi_{Wb}), 0)}$$